

Foundations of Data Analytics

Final Report

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Introduction:

In recent decades, the prevalence of obesity has surged, emerging as a global public health concern with far-reaching consequences. Concurrently, an increasing body of evidence suggests a complex relationship between Body Mass Index and physical activity. Body mass index or “BMI” is a measurement of a person's height and weight characteristics, and it categorizes them into groups, “underweight,” “normal,” “overweight,” “obese” and, “morbidly obese.” It is used to assess an individual's long-term risk of developing several health issues such as hypertension, type 2 diabetes, and coronary heart disease. Physical activity is one of the leading factors that can influence an individual's weight, hence influencing their BMI.

Regularly taking part in physical activity is crucial in the prevention of unhealthy weight gain and therefore, improving cardiovascular health. Historically, the association between BMI and physical activity has been viewed as straightforward: increased physical activity results in lower BMI and vice versa. While this concept forms the basis of many health promotion campaigns, recent studies have underscored the need for a more nuanced exploration. The nature of this correlation suggests that the relationship is not one-dimensional and may be influenced by a myriad of factors, including genetic predispositions, socio-economic variables, and cultural contexts. Studies have shown a significant inverse correlation between high physical activity levels to BMI. The aim of this cross-sectional study was to investigate whether BMI was inversely related to physical activity. Other studies have previously identified that there is a negative relationship between BMI and physical activity, while also outlining that other factor also influenced BMI. Hence, this study aims to conclude that in a group of working subjects in late adulthood, a negative association exists between BMI and physical activity.

Research Question:

The main research question of interest in this study is to investigate the indirect correlation between physical activity and an individual's body mass index. The research was carried out in a group of working subjects in late adulthood. The study also investigates whether factors such as sex, race or other socio-economic factors have an impact on an individual's BMI, regardless of physical activity levels.

Methods

The research conducted is an observational, cross-sectional study to see if there is a connection between BMI (body mass index) and physical activity. 900 subjects in late adulthood (female=0, male=1) and free from cardiovascular disease, all subject's BMI (kg/m^2) physical activity which includes any physical activity that is not work related (hours per week) were taken to analyse the correlation between the two variables, the subjects were categorized based on their BMI, “overweight “,”obese”,” morbidly obese” “normal weight”. There were no underweight subjects in this study.

Statistical analysis

Scatter and dot plots were used for initial explorational analysis of the data. Following this, a simple linear regression model was applied, and its results interpreted. Means were calculated to compare the differences in the average BMI and hours of physical activity of the male and female subjects, these were then used to determine any significant differences. All graphs and statistical analysis were done using RStudio.

Results

Initial visual inspection of the data (*Figure 1*) shows a slight negative association between physical activity and BMI. This result is within research expectations, the more physical activity a subject was engaged in, the lower the BMI was.

Furthermore, upon analysis of the simple linear regression results, it was observed that the slope of the line linked to the physical activity effect was found to be around -0.49703 . According to this, with every additional hour spent doing recreational physical activity, the BMI of the subject decreases, on average, by 0.49703 kg/m^2 .

Also, there were observable differences between males' and females' BMI which could be related back to the amount of time they each spent engaging in leisurely physical activity. As can be seen in Table 1, and Figures 3 and 4 on average where when females spent more time moving, they showed lower BMIs versus males showing higher BMIs with less time spent moving.

Figure 1. Scatterplot and line of best fit showing relationship between BMI (kg/m²) and hours of physical activity

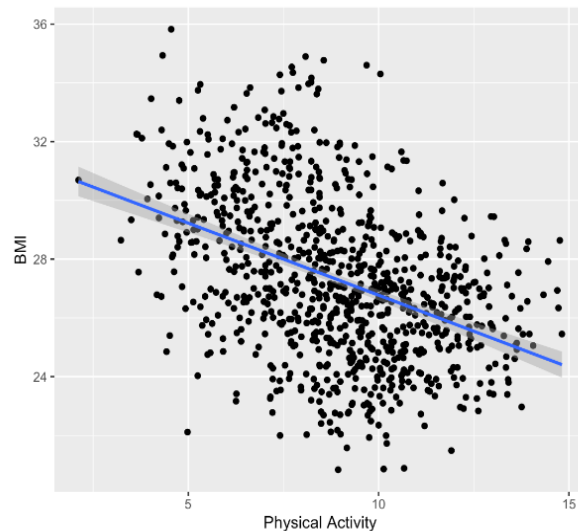


Table 1. Showing mean BMIs and time spent doing

physical activity in hours

Females (N=443)		Males (N=457)
BMI (kg/m ²)	Mean = 26.93587	Mean = 27.66199
Physical Activity (Hours)	Mean = 9.027657 hrs.	Mean = 8.806589 hrs.

Figure 3

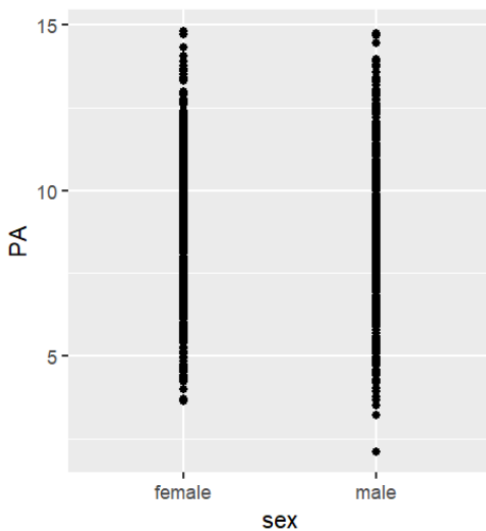
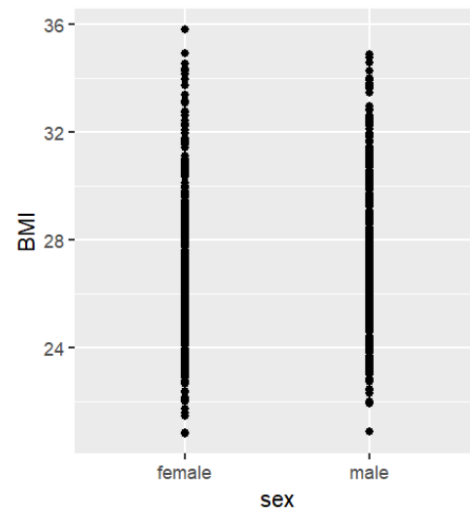


Figure 4



Additionally, t-tests conducted to determine the relationship between BMI and physical activity revealed a p-value of $2.2e-16$, suggesting that there are significant correlations between the two.

Discussion

The purpose of this study was to see the indirect/negative correlation between physical activity and BMI. The key findings of the research conducted showed that subjects with a lower BMI tend to be generally more active physically than those with a higher BMI.

It has also been previously observed that the prevalence of cardiovascular disease is higher in people who are obese/overweight and after some further inspection it can also be seen that it was more prevalent in people who did not engage in purposeful physical activity.

(Banerjee, 2021)

As can be seen in figure 1, with every hour of physical activity 0.5kg/m^2 would decrease from a person's BMI. This shows that there is a negative correlation between BMI and every hour of physical activity. There was a difference between female and males in terms of BMI, it is observed that females tended to have a lower BMI as they had more time spent doing physical activity than the male subjects.

It has also been identified that a BMI greater than 30kg/m^2 is classified as obese and it entails a higher risk of developing coronary heart disease, hypertension, some cancers, and type 2 diabetes (Chen et al., 2021).

Smoking status can also affect BMI a study has shown that the BMI of current smoker, non-smokers are different by 1.6kg/m^2 , it is seen that after long term abstinence of smoking that previous-smoker BMI return to the same as people who have never smoked (Marcus R. Munafò, 2009), this means that although smoking can lower your BMI even though it can increase risk of CVD so BMI is not the only indicator of the risk of developing cardiovascular illnesses

Limitations

Although BMI is an accessible and an easy option to calculate and determine your classification in terms of being overweight, obese, morbidly obese, normal, or overweight. BMI does not adjust differences between sex and ethnicity, nor does it consider individuals muscle mass or bone density. BMI also does not account for fat distribution in the body. The way fat is stored and distributed in the body can have an impact on an individual's BMI. BMI does not account for this difference.

BMI can mean high levels of body fat, but it could also mean higher muscle mass. Males tend to have a higher muscle mass than females. As a result, males usually have a higher BMI than females. This does not mean that a male with a higher BMI than a female is 'unhealthier'.

Considering the study is done on subjects in late adulthood we cannot see or analyze the effects of physical activity on BMI in younger people as they tend to have a faster metabolism and tend to have much higher levels of deliberate recreational physical activity. Another factor also not taken into consideration is socio-economic and job status as this largely determines the type and quality of food they consume. This factor also ties in with the amount of time they spend moving about as some people may be too busy to exercise or alternatively have even more time on their hands.

Implications

The outcome of this research further proves the initial hypothesis of the study that spending more time engaging in physical activity leads to healthier BMIs. This outcome will significantly help in healthcare as this research will encourage others to engage in regular leisure-time physical activity leading to obesity prevention. In addition, this research could be used to explore more specific methods behind the relationship BMI and physical activity for better treatments and interventions and healthcare providers can also assess the health risks and modify personalized health plans.

Referencing

- Banerjee, S. A. K., Pradeep AND Srivastava, Shobhit AND Banerjee, Adrita},. (2021). Association of anthropometric measures of obesity and physical activity with cardio-vascular diseases among older adults: Evidence from a cross-sectional survey, 2017–18},. *PLOS ONE* 16, 1-16.
- Chen, Y.-T., Lee, P.-F., Lin, C.-F., Chang, A., Chung, Y.-C., Yeh, C.-P., Chen, S.-C., & Ho, C.-C. (2021). Association of Regular Leisure-Time Physical Activity with Self-Reported Body Mass Index and Obesity Risk among Middle-Aged and Older Adults in Taiwan. *Healthcare*, 9(12), 1719.
<https://www.mdpi.com/2227-9032/9/12/1719>

Marcus R. Munafò, K. T., Yoav Ben-Shlomo. (2009). Smoking status and body mass index: A longitudinal study Get access Arrow. *Nicotine & Tobacco Research*, 11(6), 765-771.